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Multi-objective federated learning: Balancing global performance and individual fairness

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ABSTRACT

In federated learning, non-iid data not only diminishes the performance of the global model but also gives rise to the fairness problem which manifests as an increase in the variance of the global model's accuracy across clients. Fairness issues can result in the global model performing poorly or even failing on certain clients. Existing methods addressing the fairness problem in federated learning tend to neglect the comprehensive improvement of both the average performance and fairness of the global model. In addressing it, the multi-objective optimization method for fine-tuning global gradients, FedMC algorithm is introduced in this paper. The primary objective is the average loss function of all clients, and the sub-objective involves fine-tuning the global gradient by reducing the gradient conflict between the global gradient and the local gradients. Specifically, we refine the global gradient by incorporating a sub-optimization objective aimed at alleviating conflicts between the global gradient and the local gradient with the largest deviation, denoted as FedMC. FedMC can enhance the performance and convergence rate of clients with initially poor performance, albeit at the cost of the earlier convergence rate of clients with initially good performance. Nevertheless, it enables the latter to reach the accuracy level achieved before fine-tuning. In addition, we also propose FedMC+ algorithm, owning three additional optimization mechanisms built upon the FedMC optimization objective which includes the decay of hyperparameter, the sliding window mechanism, and data-balanced client selection. Besides, we present a theoretical analysis of the convergence rate of FedMC, demonstrating its convergence to a Pareto stationary solution. Our combined experimental results confirm that FedMC+ achieves an average 4.5% improvement in accuracy and a 22% reduction in the degree of dispersion compared to state-of-the-art federated learning (FL) methods.

1. Introduction

Federated learning represents a distinctive form of distributed machine learning system, characterized by an extensive network of remote devices with privacy protection [1–5]. Confronted with the diversity of devices, federated learning effectively addresses the challenge of reconciling the model training need for substantial dataset with the limitations of the server's data collection capabilities [6–8]. However, the challenge hindering the improvement of federated learning (FL) model performance primarily stems from the statistical-level imbalance in data labels and features [9–15]. The impact of non-iid data on the global model is typically comprehensive, leading to a decline in multiple indicators. Notably, the non-iid data in FL inherently leads to model performance degradation, slowed convergence, and fairness challenges at a statistical level. Fairness, which is often optimized in computer networks[16] and dispatching systems[17], is a reasonable expectation from clients for global models, meaning that every client anticipates the global model to exhibit good performance on their respective local datasets. It pertains to the consistency of the global model's performance distribution on each client which is often overlooked in federated learning. The fairness can be quantified by measuring the variance of the global model's accuracy across local datasets.

The existence of a single global model makes it difficult to take care of the performance on each client, especially in FL scenarios with heterogeneous data distribution which causes the problem of unfairness. To address the problem of FL fairness, existing approaches were proposed from different perspectives. AFL [18] is proposed to focus on protecting the worst-case performance on all clients. Li et al. [19] modifies the global objective by using fair resource allocation approach to balance the performance and fairness. Fedmdga+ [20] improves fairness and robustness without sacrificing model performance by designing a multi-objective optimization problem. Ditto [21] measures

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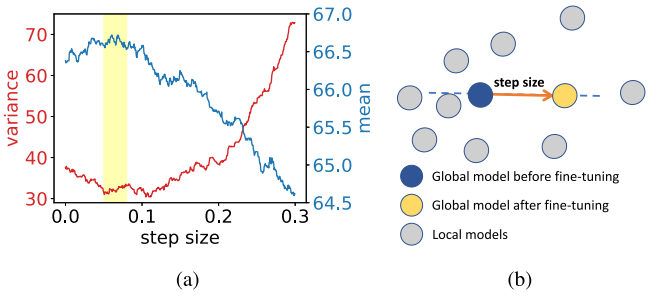


Fig. 1. The trend of global model's average accuracy and accuracy variance on local data set with the deviation step size increasing. (a) plot of the trend between the accuracy variance (and mean) and the global model deviation step size; (b) diagram of global model deviation.

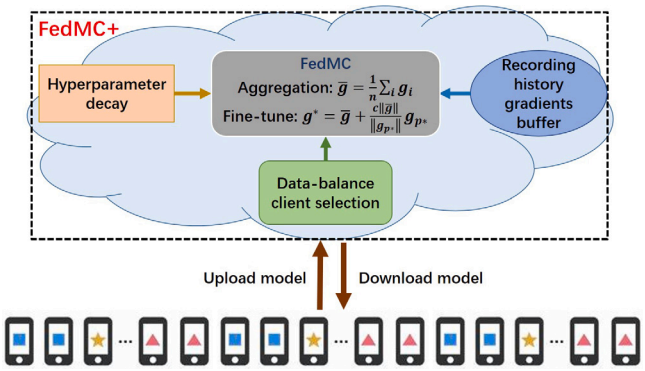


Fig. 2. The frame diagram of FedMC and FedMC+.

the trade-off between robustness and fairness based on the design of a personalized model. Wang et al. [22] propose that gradient conflict is one aspect of affecting federated learning fairness, and use gradient surgery to reduce the conflict of each local gradient. The existing federated learning fairness methods often resort to rudimentary modifications of the global loss function or the design of sub-optimization goals. Unfortunately, these approaches often lack a clear definition of primary and secondary objective, or simply address the fairness of the model by adding specific mechanisms and the sacrifice in the pursuit of fairness are not explicitly clarified.

During the training process of federated learning, if we design the global model to shift in the direction of the local model far away from it, we find that the global model has higher test accuracy and lower accuracy variance within a certain step size range. Fig. 1 illustrates the variation in accuracy mean and accuracy variance resulting from the step size of the global model's shift towards a local model that is significantly distant from it at the 50th global round of training. Building upon these findings, we posit that fine-tuning the global model to shift towards a local model significantly distant from it can yield a fairer global model and enhance overall model performance. From this point of view, a modest fine-tuning of the global model within a certain neighborhood can balance the distribution of the global model's performance on each client dataset.

In our research, we observe that if the global gradient aligns too closely with the local gradient in the dominant direction, the global gradient may fail to capture the rich information contained in the local gradient. This, in turn, can lead to fairness issues. Indeed, the higher the similarity between gradients, the more shared information they encompass. As a result, we devise an easily fine-tuned multi-objective optimization to alleviate conflicts between the global gradient and specific partial local gradients. In this fine-tuned multi-objective framework, the primary objective is established as the common global optimization objective in federated learning, specifically the expected empirical loss across all client datasets. The sub-objective is set as alleviating the gradient conflict between the global gradient and the local gradients with constraint condition. This approach aims to extract the latent gradient information embedded in these local gradients without impacting the performance of the global model. This enables the fine-tuned global gradient to generate a more universally applicable global model. We introduce FedMC to address the sub-objective of alleviating conflict between the global gradient and local gradients by fine-tuning the global gradient based on the optimal global loss function of the primary objective. Gradient conflict not only leads to significant differences between local gradients, hindering the overall performance of the global gradient but also gives rise to fairness issue due to the prevalence of dominant local gradients. We refine the global gradient by formulating a convex optimization problem aimed at minimizing the difference between the global gradient and its most diverse local gradient which does not compromise the accuracy of the well-performing clients when it eventually converges.

We introduce FedMC+ as a solution to the challenges posed by non-iid data in federated learning, addressing issues such as global model performance degradation and uneven distribution of local performance. The framework diagram of FedMC and FedMC+ is shown in Fig. 2. FedMC+ comprises a foundational algorithm, FedMC, complemented by three optimization mechanisms. The first mechanism used is the decay of hyperparameter c, which controls the search space of the convex optimization problem. Secondly, a sliding window mechanism is employed to augment the diversity of local gradients introduced by the algorithm. Finally, the data balanced client selection is implemented to alleviate global gradient conflicts over time series.

The contributions of our work are as follows:

- We introduce FedMC as a method to mitigate model performance degradation and uneven model performance distribution in non-iid federated learning. Building upon FedMC, we present FedMC+ as the enhanced version, which incorporates three additional optimization mechanisms.
- We conduct a rigorous theoretical analysis of the convergence of FedMC in non-iid scenario and demonstrate that FedMC converges to a Pareto stationary point.
- The advantages of FedMC+ are validated over state-of-the-art methods by conducting comprehensive evaluations on multiple federated learning datasets, considering accuracy, fairness, and convergence rate.

The structure of the paper is outlined as follows. Section 2 provides an overview of related work in federated learning (FL) addressing issues of cooperative fairness and performance degradation in non-iid scenarios. Section 3 delves into the preliminaries of FL. In Section 4, we provide a detailed introduction to the proposed algorithm. Subsequently, Section 5 is dedicated to the experimental evaluation and comparison of the proposed algorithm. The paper concludes in Section 6, summarizing the key findings and contributions.

2. Related work

The classical Fedavg algorithm has good universality in many applications, and its convergence under iid data has been proved [23]. However, under the setting of non-iid, the significant difference of data distribution becomes a momentous factor to hinder the convergence of the global model. If the number of local update steps increases, the overfitting of the local model leads to the oscillation of global model convergence [24]. Zhao et al. [15] show that the attenuation of federated learning convergence on account of the non-independently identical data is proportional to the number of local update steps. The convergence of federated learning under non-independently identical

distribution is also analyzed by Li et al. [25], where they propose a bound that demonstrates a trade-off between communication efficiency and convergence rate.

In the optimization method to reduce the influence of non-iid on the model preformance, the basic idea is to modify the local training process or the global aggregation process so that the global model is closer to the real centralized model. FedProx algorithm [26] typically modifies the loss function of the local model and adds a regular term for the distance between the local model and the global model of the previous round to prevent the convergence oscillation of the global model caused by large differences between the local models trained by each client. FedDyn algorithm [27] makes the fixed point of the local loss function consistent with that of the global loss function by adding dynamic regular terms. FedDF algorithm [28] adds iterative training of knowledge distillation to the global model on the server side, while FD algorithm [29] adds penalty term of knowledge distillation to local training on the client side. Both methods improve the robustness of model training of federated learning under non-iid distribution through knowledge distillation. MAFL [30] algorithm is that each client sends or accepts the local average data of other clients to supplement its local data to reduce the degree of non-iid data. The FedBE algorithm proposed by Chen et al. [31] introduces the idea of integrated learning from the statistical level of Bayesian inference and combines the method of knowledge distillation to solve the problem of non-iid. The SCAFFOLD algorithm proposed by Sai Praneeth Karimireddy et al. [24] measures the "client-side drift" of non-iid data distribution and eliminates the "client-side drift" through iteration during local user training. Li et al. [32] proposed the algorithm MOON that utilizes the similarity between model representations to correct local training by adding contrast loss functions.

The different needs of federated learning have different definitions of fairness. In general, there are two definitions of fairness: collaborative fairness and equilibrium fairness. Some researches focus on measuring the contribution of each client and re-assigning weight to them to achieve collaborative fairness [33,34]. Li et al. [18] proposed AFL aiming to improve the worst-case performance of any client to achieve fairness. The q-FEL and q-Fedavg algorithms proposed by Li et al. [19] introduce the problem of fair allocation of model resources to each user in non-iid scenarios and introduce the hyperparameter q into the global loss function, which enables the global model to have good performance on each local data set. Hu et al. [20] leverages the Pareto optimal method in multi-objective optimization to propose Fedmdga+ to ensure fairness and robustness. Wang et al. [22] introduce the gradient conflict of FL and solve the fairness problem by solving the internal and external gradient conflict. FedABC [35] algorithm takes into account unfair competition issue caused by the imbalanced data distribution and lack of positive samples for some classes in each client. Ye et al. [36] propose a preference-aware scheme in Fair Federated Learning paradigm to adaptively adjust the model based on each client's preferences to meet their needs.

As a solution to model performance degradation caused by data heterogeneous environment, personalized federated learning can customize personalized model for each client so that each personalized model can adapt to its corresponding local data set. However, fairness is a concept for global model, which describes whether a global model can fairly allocate its model performance resources to each client. In comparison, a fairer global model will be more generalized. In personalized federated learning, FedFomo [37] updates the aggregate weights of each client personalization model by optimizing convex optimization problems. Similarly, DoFed-SPP [38] updates the weights of personalized model aggregation by designing convex optimization objectives, which solve semi-supervised federated learning scenarios under the constraints of network transmission and training time. MiniPFL [39] is a bi-component framework that selectively prioritizes valuable clients for optimizaing personalized federated learning. Ditto [21] addresses both fairness and robustness through a simple personalized federated learning update framework. FedAMP [40] calculates the importance of other local models to the aggregation of the personalized model by introducing an attentive message passing mechanism. APPLE [41] is a personalized cross-silo FL framework that adaptively learns how much each client can benefit from other clients' models. FedALA [42] captures the global model information required by the personalized model by dynamically updating the aggregate weights of the personalized model.

3. Preliminaries

We explicitly describe the symbols used for federated learning in this section. Suppose each client has its own dataset D_i. The federated learning goal is to learn a shared parameter w over union dataset $D = \bigcup_{i\in[N]} D_i$. Translating it into an optimization objective is to minimize the weighted average loss of all clients:

$$\min_{w\in\mathbb{R}^n} F(w) \triangleq \sum_{i=1}^{N} p_i f_i(w) \tag{1}$$

where $f_i(w)$ is the ith local objective function, N is the number of participating devices and p_i is the weight of the ith client satisfying $p_i \geq 0$ and $\sum_{i\in[N]} p_i = 1$. We define n_k as the number of local data samples of the kth device, and p_k can be calculated as $n_k / \sum_{i\in[N]} n_i$. The local objective is often defined as the empirical risk on the local dataset, *i.e.* $f_i(w) = \frac{1}{|D_i|} \sum_{(x_i,y_i)\in D_i} l(w;(x_i,y_i))$ where $l(;)$ is the cross-entropy loss.

As for the fairness in federated learning, we refer specifically to equilibrium fairness. For a global model w, its performance indicator on the ith client dataset is denoted as $\mathcal{M}_i(w)$ (accuracy or loss). Then the variance of the performance indicator $Var[\mathcal{M}_i(w)]$ can measure the fairness of the global model w on each client. We consider the global model w is fairer than w' if they satisfy $Var[\mathcal{M}_i(w)] < Var[\mathcal{M}_i(w')]$.

4. Mitigate gradient conflicts to refine fairness

In this section, we design a multi-objective optimization to reduce gradient conflict which to a large extent hinders the fairness of the performance of FL model and call its solution algorithm FedMC. FedMC aims to fine-tune the global model to promote fairness without compromising model performance. FedMC+ has one main optimization goal and three mechanisms for mitigating gradient conflict (parameter decay, sliding window, and data balanced client selection). In addition, the theoretical analysis of the convergence of FedMC and how the algorithm converges to the Pareto optimum is also provided.

4.1. Objective of mitigating gradient conflict

Due to the heterogeneity and imbalance of data in federated learning, there are great differences between the gradients obtained from the data of each client. The global gradient generated by the aggregation of these gradients with large differences has a high probability of conflict with some of the local gradients, where such conflict is defined that if $\langle g_1, g_2\rangle < 0$, then we say that gradient g_1 and g_2 are in conflict. If the conflict between the global gradient and the local gradient is too large, the performance of the global model in the local task deteriorates or even becomes invalid. Therefore, it is not only necessary to ensure the accuracy of the global model from the "macro perspective" but also to prevent the performance failure of the global model on some local tasks from the "micro perspective" in federated learning.

In the paper, we use the inner product $\langle\cdot,\cdot\rangle$ of gradients to define the gradient difference where the greater the inner product, the smaller the gradient difference. Therefore, we need to find a global update direction that converges to the optimal value of the global expected loss while reducing the gradient conflict. We refer to the optimization

objective of solving gradient conflict in multi-task learning in [43], which is designed as follows:

$$\max_{g\in\mathbb{R}^n}\min_{i\in[K]}\langle g_i, g\rangle \quad s.t. \quad \|g-\bar{g}\|\le c\|\bar{g}\|, \tag{2}$$

where g is the global gradient that can mitigate the gradient conflict, $\bar{g}$ is the baseline gradient which can be aggregated from any federated learning strategy and c is the hyperparameter used to control the search range of g. The optimization goal (2) is to find the local gradient that is the most differentiated from the global gradient, and then minimize its differences. At the same time, in order to avoid excessive deviation from the main optimization goal of federated learning, we control the search range of the variables of the optimization goal around the gradient aggregated by some federated learning strategy.

In order to transform Problem (2) into convex optimization mode, we introduce a set of convex combinations that transform the original optimization variables into the coefficients of the convex combination. Specifically, $min_i\langle g_i, g\rangle = min_{p\in\mathcal{P}}\langle\sum_i p_i g_i, g\rangle$, where $p=(p_1,p_2,\dots,p_N)\in\mathbb{R}^N$ and $\mathcal{P}$ denotes the probability simplex, *i.e.* $\mathcal{P}=\{p : \sum_i p_i = 1 \text{ and } p_i\ge 0\}$. We denote $g_p=\sum_i p_i g_i$ and $\phi = c^2\|\bar{g}\|^2$. Then the Lagrangian of the objective (2) is:

$$\max_{g\in\mathbb{R}^n}\min_{\lambda\ge 0, p\in\mathcal{P}} g_p^T g - \frac{\lambda}{2}(\|g-\bar{g}\|^2-\phi), \tag{3}$$

where λ is Lagrange multiplier. Since the primal problem (3) is concave with g (due to negative quadratic form) and satisfies Slater's condition, we can obtain its dual problem by swapping *max* and *min*:

$$\min_{\lambda\ge 0, p\in\mathcal{P}}\max_{g\in\mathbb{R}^n} g_p^T g - \frac{\lambda}{2}(\|g-\bar{g}\|^2-\phi). \tag{4}$$

Fixing λ and p, and setting the derivative of problem (4) with respect to g equal to zero yields $g=\bar{g}+\frac{g_p}{\lambda}$, which is the minimum point of problem (4). Then by applying g we can write (4) as:

$$\min_{\lambda\ge 0, p\in\mathcal{P}} g_p^T\bar{g}+\frac{1}{2\lambda}\|g_p\|^2+\frac{\lambda\phi}{2}. \tag{5}$$

By applying AM-GM inequality, we have

$$\begin{aligned} &g_p^T\bar{g}+\frac{1}{2\lambda}\|g_p\|^2+\frac{\lambda\phi}{2}\\ \ge& g_p^T\bar{g}+2\sqrt{\frac{1}{2\lambda}\|g_p\|^2\cdot\frac{\lambda\phi}{2}}\\ =& g_p^T\bar{g}+\sqrt{\phi}\|g_p\|, \end{aligned} \tag{6}$$

when $\frac{1}{2\lambda}\|g_p\|^2=\frac{\lambda\phi}{2}$, inequality (6) takes an equal sign. So if we take $\lambda=\|g_p\|/\sqrt{\phi}$, then problem (5) is transformed into the following problem:

$$\min_{p\in\mathcal{P}} g_p^T\bar{g}+\sqrt{\phi}\|g_p\|. \tag{7}$$

Since the optimal solution of problem (2) is $g^*=\bar{g}+\frac{\sqrt{\phi}}{\|g_{p^*}\|}g_{p^*}$, where $g_{p^*}=\sum_i p_i^* g_i$ and p^* is the optimal solution of (7), g^* can be regarded as the modified gradient obtained by shifting towards g_p based on $\bar{g}$, making g^* more equitable in alleviating gradient conflict. Thus, g^* is used as the direction of the global update on the FedMC server. As for calculating g_p^* which is a convex combination of g_i, the optimization problem (7) is a linear optimization problem that can be solved by the optimizer in cvxpy, a third-party library for solving convex optimization problems in python. The following introduces three mechanisms which are separately hyperparameter decay, sliding window, and data balanced client selection on FedMC which is called FedMC+ and described in Algorithm 1.

The adjustment of the global gradient is beneficial in enhancing the accuracy of clients with poor performance, as the global gradient direction is adjusted toward the direction of the local gradient for these clients. However, this adjustment causes the global gradient to deviate from the local gradient of well-performing clients. Nonetheless, as the adjusted global gradient direction remains the acute angle with the local gradient direction associated with well-performing clients, the primary impact is a reduction in the convergence speed of these clients in the early stages. Yet, with a sufficient number of training rounds, the accuracy of these clients can still reach the level achieved before fine-tuning the global gradient.

Algorithm 1 FedMC+ (Federated Mitigating conflict)

Input: K, T, η_g, c, ω, N.
Initialization: Randomly initialize global model w_0 and initialize gradient record GR as an empty list.
Output: Global model w_{T-1}.
Server computes A and b by collecting each client data category distribution.
for $t=0$ to $T-1$ **do**
  Select client set $S_t \leftarrow DBS(A,b,K,N)$.
  Send w_t to S_t.
  for on client $i\in S_t$ parallel do **do**
    Receive w_t.
    Conduct Local Update with objective function f_i.
    Send local update g_i^t.
  end for
  Receive local updates $[g_i^t]_{i\in S_t}$.
  Update gradient record: $GR \leftarrow UGR(GR, [g_i^t]_{i\in S_t}, \omega)$.
  Calculate size of GR as L and compute baseline global gradient $\bar{g}_t=\frac{1}{L}\sum_{i\in S_t} g_i^t$.
  Get optimal weight $\{p^{*t}_i\}_{i=1}^L$ by addressing the optimization objective (7).
  Calculate $g_{p_t^*}=\sum_{i\in[L]} p^{*t}_i\cdot\hat{g}_i^t,\ \hat{g}_i^t\in GR$.
  Global update: $w_{t+1}=w_t-\eta_g(\bar{g}_t+\frac{c\|\bar{g}_t\|}{\|g_{p_t^*}\|}g_{p_t^*})$.
  Attenuate parameter c by some strategy.
end for

4.2. Theoretical analysis

FedMC has two convergence rates depending on the range of c ($0\le c<1$ and $c>1$). Before presenting the two convergence rates, we make the following assumptions about the loss function in federated learning.

Assumption 1. Assume that the global loss function $F(w)$ and the local loss function $f_1(w),\ f_2(w),\dots,\ f_N(w)$ are differentiable on $\mathbb{R}^m$ and are separately L-smooth:

$$F(w_1)\le F(w_2)+(w_1-w_2)^T\nabla F(w_2)+\frac{L}{2}\|w_1-w_2\|^2,$$
$$f_i(w_1)\le f_i(w_2)+(w_1-w_2)^T\nabla f_i(w_2)+\frac{L_1}{2}\|w_1-w_2\|^2.$$

Their gradients $\nabla F(w)$ and $\nabla f_i(w)$ are separately L-Lipschitz and L_1-Lipschitz:

$$\|\nabla F(w_1)-\nabla F(w_2)\|\le L\|w_1-w_2\|,$$
$$\|\nabla f_i(w_1)-\nabla f_i(w_2)\|\le L_1\|w_1-w_2\|,$$

where $i\in[N]$ and $L,\ L_1$ are positive constants. In addition, $F^*=inf_{w\in\mathbb{R}^m}F(w)>-\infty$.

Assumption 2. Let ξ_i^t be sampled from ith device's local data uniformly at random. The variance of stochastic gradients in each device is bounded:

$$\mathbb{E}\|\nabla f_i(w_t,\xi_i^t)-\nabla f_i(w_t)\|^2\le\sigma_i^2.$$

for $i=1,2,\dots,N$. If the global gradient is defined as the average of the local full gradients, *i.e.* $\nabla F(w_t)=\frac{1}{N}\sum_{i=1}^N\nabla f_i(w_t)$ and the estimated global gradient is defined as the average of the local gradients by stochastic gradient descent (SGD), *i.e.* $\bar{g}_t=\frac{1}{N}\sum_{i=1}^N\nabla f_i(w_t,\xi_i^t)$ which

satisfies $\nabla F(w_t) = \mathbb{E}\bar{g}_t$, then the variance of global gradient is bounded as:

$$\mathbb{E}\|\nabla F(w_t) - \bar{g}_t\|^2 \leq \frac{1}{N}\sum_{i=1}^{N}\sigma_i^2.$$

See [25] for the relevant proof of the variance bound of the global gradient.

Assumption 3. The expected norm of stochastic gradient is uniformly bounded, *i.e.* $\mathbb{E}\|\nabla f_i(w_t^i, \xi_t^i)\| \leq G$ for all $i = 1, \ldots, N$ and $t = 0, \ldots, T+1$. The difference between the second order origin moment of the averaged stochastic gradient and the global gradient is also bounded, *i.e.* $\mathbb{E}\|\bar{g}_t\|^2 \leq \|\nabla F(w_t)\|^2 + \sigma^2$ where $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}\sigma_i^2$. The above two inequalities are for all $i = 1, \ldots, N$ and $t = 0, \ldots, T+1$.

Based on the assumptions above, we present convergence rate of FedMC when $0 \leq c < 1$ in the federated learning scenario with non-iid data.

Theorem 1. *Assume Assumptions 1, 2 and 3 hold. With a fixed learning rate α satisfying $0 < \alpha < \frac{1}{L}$ and c satisfying $0 \leq c < 1$, FedMC converges to stationary points of F with the convergence rate*

$$\frac{1}{T+1}\sum_{t=0}^{T}\|\nabla F(w_t)\|^2 \leq \frac{2(F(w_0) - F^*)}{\alpha(1-c^2)(T+1)} + \frac{\sigma^2 + 2c\sigma G + c^2\sigma^2}{1-c^2},$$

where $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}\sigma_i^2$ and G is the bound of the expected norm of stochastic gradient.

Please see Appendix A for relevant proof. The following shows an additional convergence rate when $c > 1$. The following proof assumes a proper decay step size, and the fixed points of FedMC are either stationary points of F or Pareto-stationary points of $(f_1, \ldots, f_N)$.

Theorem 2. *Assume Assumptions 1, 2 and 3 hold. With $c > 1$ and a time varying step size satisfying*

$$\alpha_t \leq \frac{\|\bar{g}_t\|}{L_1(c-1)\|g_{p_t^*}\|},$$

we have:

$$\sum_{t=0}^{T}\alpha_t\|\bar{g}_t\|\|g_{p_t^*}\| \leq 2\frac{min_i(f_i(w_0) - f_i(w_{T+1}))}{c-1},$$

where p_t^* is the solution of (7) at the t-th iteration. Therefore, if we have $f_i^* = inf_{w\in\mathbb{R}^m}F(w) > -\infty$ and $c > 1$, the series on the left-hand side of the inequality converges, meaning that either $\alpha_t \to 0$, or $\|\nabla F(w_t)\| \to 0$, or $\|g_{p_t^*}\| \to 0$. Then the actual behavior of the algorithm depends on the specific choice of the step size. For example, if we take $\alpha_t = \frac{\|g_{p_t^*}\|}{L_1(c-1)\|\bar{g}_t\|}$, the results become:

$$\sum_{t=0}^{T}\|\nabla F(w_t)\|^2 \leq 2L_1 min_i(f_i(w_0) - f_i(w_{T+1})),$$

which ensures $\|\nabla F(w_t)\| \to 0$.

Please see Appendix A for relevant proof. Considering the fairness of federated learning, we introduce the concept of Pareto optimality, which expresses the strategy that each individual can achieve the optimal allocation of resources when the overall model resources are certain. Here we introduce a definition of Pareto optimality, which can be easily applied to determine whether it is Pareto optimal.

Definition 1. For smooth criteria $f_i(w)$ $(1 \leq i \leq N)$. w^0 is called a Pareto-stationary if there exists some convex combination of the gradients $\nabla f_i(w^0)$ that equals zero [22].

Theorem 3. *For any $c \geq 0$, all the fixed points of FedMC are Pareto-stationary points of $(f_1, \ldots, f_N)$.*

Proof. For general $c \geq 0$, in the fixed point, we have $d^*(w) = \bar{g} + \lambda g_{p^*} = 0$, then we have

$$\sum_{i=1}^{N}\frac{\frac{1}{N} + \lambda p_i}{1+\lambda}g_i = 0,$$

where λ is a positive constant, $\frac{\frac{1}{N}+\lambda p_i^*}{1+\lambda} \geq 0$ and $\sum_{i=1}^{N}\frac{\frac{1}{N}+\lambda p_i^*}{1+\lambda} = 1$. Therefore there exists some convex combination $\{\frac{\frac{1}{N}+\lambda p_i^*}{1+\lambda}\}_{i=1}^{N}$ of the gradients $\{g_i\}_{i=1}^{N}$ that equals zero. □

4.3. Examination

In this section, we aim to propose a decision condition for assessing the appropriateness of the degree of fine-tuning applied to the global gradient. In general, our goal is to fine-tune the global gradient direction in such a way that each local empirical loss is advantageous, meaning that every local empirical loss function exhibits a decreasing trend. While adjusting the direction of the global gradient toward the local gradient dissimilar to it, it is crucial to ensure that the angle between the local gradient in the original dominant direction and the new global gradient is not obtuse, specifically satisfying the condition: $g^{*T} \cdot g_{dom} > 0$, where $g_{dom} = \max_{i\in[N]}(\bar{g}^T \cdot g_i)$.

Then, by considering the upper bound of the loss function at each step, we have the flexibility to adjust the hyperparameter c to ensure that the upper bound at each step consistently remains less than 0. When $0 < c \leq 1$, we can infer from Theorem 1:

$$\begin{aligned}
&F(w_{t+1}) - F(w_t)\\
&\leq -\frac{\alpha}{2}(1-c^2)\|\nabla F(w_t)\|^2 + \frac{\alpha(\sigma^2 + 2c\sigma G + c^2\sigma^2)}{2}\\
&= \frac{\alpha\|\nabla F(w_t)\|^2 + \alpha\sigma^2}{2}c^2 + \sigma G c + \frac{\alpha\sigma^2 - \alpha\|\nabla F(w_t)\|^2}{2}.
\end{aligned}$$

To guarantee that the upper bound above possesses solutions less than 0 within the range where $0 \leq c < 1$, it is essential to ensure that the overall upper bound above is less than 0 when $c = 0$. Then we have $\sigma^2 < \|\nabla F(w_t)\|^2$, which aligns with intuition, when the degree of dispersion of the local gradient is excessively large, finding a suitable c to diminish the global loss function becomes difficult. When $c > 1$, we have from Theorem 2:

$$\begin{aligned}
&f_i(w_{t+1}, \xi) - f_i(w_t, \xi)\\
&\leq -\alpha_t(c-1)\|\bar{g}_t\| \cdot \|g_{p_t^*}\| + \frac{L_1\alpha_t^2}{2}(c-1)^2\|\bar{g}_t\|^2\\
&\leq \alpha_t\|\bar{g}_t\|(c-1)\cdot\left(\frac{L_1\alpha_t\|\bar{g}_t\|}{2}c - \frac{L_1\alpha_t\|\bar{g}_t\|}{2} - \|g_{p_t^*}\|\right)
\end{aligned}$$

We know that the above upper bound has two zeros with respect to c, $c = 1$ and $c = 1 + \frac{2\|g_{p_t^*}\|}{L_1\alpha_t\|\bar{g}_t\|}$. Therefore, when $1 < c < 1 + \frac{2\|g_{p_t^*}\|}{L_1\alpha_t\|\bar{g}_t\|}$, it is possible to make each local experience loss function decrease at each step. Furthermore, by ensuring that the aforementioned upper bound is less than zero, we can identify the range of α_t for $\alpha_t < \frac{2\|g_{p_t^*}\|}{L_1(c-1)\|\bar{g}_t\|}$ which is more lenient compared to the condition for α_t specified in Theorem 2.

4.4. Decay of hyperparameter c

The hyperparameter c controls the search range of g in problem (2), where the larger c is, the wider the search range of g is. Here we make a qualitative discussion on how c influences the optimization results. We rewrite the problem (7) as:

$$\min_{p\in\mathcal{P}} \|g_p\| \cdot \|\bar{g}\| \cdot (\cos\langle g_p, \bar{g}\rangle + c), \tag{8}$$

where $\cos\langle g_p, \bar{g}\rangle$ is the cosine of g_p and $\bar{g}$. When $0 < c < 1$, as long as $\cos\langle g_p, \bar{g}\rangle < c$, then the minimum of the problem (8) must be less than 0, so the direction of g_p^* must be obtuse from the direction of $\bar{g}$. The optimization results in this case are in line with mitigating the conflict between local gradients and global gradient. However, when

c is greater than 1, the minimum of the problem (8) is greater than 0 anyway, and its minimum depends on two factors: $\cos\langle g_p, \bar{g}\rangle$ and $\|g_p\|$. When $\|g_p\|$ dominates the result of problem (8), g_{p^*} does not necessarily conflict with $\bar{g}$, but has a smaller norm value which is not beneficial to ease gradient conflict and promote convergence. The oscillation and decay appear in the model convergence curve when $c > 1$, which has been verified in the experiment. However, when the model starts training, a larger search space can give the model greater perturbation and randomness, which makes it easier to jump out of the local optimal.

Therefore, we introduce the decay mechanism of c in the design of the algorithm, so that it decays from the number greater than 1 to the number greater than 0 and less than 1 with the global rounds. At the beginning of training, we set c slightly larger to leave a large search space for g in the problem (2). When the global model is about to converge, the search space of g should be reduced to mainly solve the conflict between local gradients and global gradient.

4.5. Sliding window mechanism

In FL, the communication cost of delivering the model is too high, so at the end of each training round, the server randomly selects partial clients to upload local models to reduce the communication cost. However, the gradient information of the partial clients inevitably deviates from the global gradient information, so we set a sliding window to use the local gradients selected in the previous rounds to reduce the deviation of the partial local gradients from the global gradient information estimation. Specifically, in problem (2), the range of g_i should belong to the set consisting of the local gradients selected in the current round and the gradients saved in the sliding window. In order to ensure the timeliness of the gradient stored in the sliding window, we specify that the sliding window is a "front-out last-in" queue of size ω, which is described in Algorithm 2.

Algorithm 2 UpdateGR: Update Gradient Record (UGR)

Input: GR, $[g_i^t]_{i\in S_t}$, ω.
Merge GR and $[g_i^t]_{i\in S_t}$:
$GR \leftarrow GR + [g_i^t]_{i\in S_t}$
Calculate the length of GR as L.
if $L > \omega$ **then**
  $GR \leftarrow GR[L-\omega :]$
end if
return GR

Algorithm 3 Data Balance Selection (DBS)

Input: A , b, K, N.
Random generate an integer (or some integers) $r \in [N]$ as the required client(s).
Optimize objective (9) with $x[r] = 1$ and obtain x^*.
Generate the selected clients set:
$S \leftarrow \{i | x^*[i] = 1,\ i \in [N]\}$
return S

4.6. Data-balanced client selection

Random client selection is easy to cause unbalanced selection data, especially in extremely non-iid case, where the dominant data category is different in each round. The alternating initialization and training of each round cause a pull in the direction of the gradient, resulting in a large oscillation of the convergence curve, which is called gradient conflict on the time series. Therefore, we design a client selection method to maintain data balanced so that the selected local gradients of each round contain various categories of data information to reduce the pulling of each round in the direction of the gradient.

We denote $A \in \mathbb{R}^{C\times N}$ as the matrix of the number of each data category per client where C is the number of data categories. We define $x \in \mathbb{R}^N$ as a selection vector, which is a binary vector consisting of 0–1, where 0 means not selected and 1 means selected. $b \in \mathbb{R}^C$ is defined as the amount of data for each data category selected in each round, where each element in b is set to $\frac{D}{KC}$ in order to get balanced data. Then, as long as the equation $Ax = b$ is satisfied, the data selected in each round is balanced. Generally speaking, since $N \gg C$, the column vectors of A are linearly dependent with a high probability, then the linear system consists of an infinite number of solutions. Therefore, we design the following optimization objective to solve the choice vector x:

$$\begin{aligned} &\min_{x\in\mathbb{R}^N} \|Ax - b\|^2 \\ &s.t. \quad \mathbf{1}^T x = K, \\ &\qquad x[i] = 0 \ or \ 1, \end{aligned} \tag{9}$$

where $i = 1, 2, \ldots, N$ and $\mathbf{1}$ is a column vector where every element is 1. The client selection method does not occupy too much communication overhead. It only needs each client to upload its own vector composed of the number of each data category to the server before training. The additional communication overhead of this operation is $O(N \times C)$.

Since both A and b are fixed, the optimal vector x^* obtained in each round is also fixed. In order to ensure that every client has the opportunity to participate in aggregation, we randomly select a part of the client $\hat{S}$ (where $\hat{S} \subset S$ and $|\hat{S}| = K_1$) as a required option in each round of client selection, and then optimize the remaining $K-K_1$ clients to be selected by solve problem (9). Data-balanced client selection is described in the algorithm 3.

5. Experiment

In this section, we empirically evaluate the proposed FedMC with its three mechanisms compared to different FL benchmarks on various scenarios. In terms of experimental indicators, the convergence curves of the FedMC and FedMC+ algorithms are first compared with those of other algorithms. This comparison allows for the evaluation of basic performance metrics, such as convergence rate and convergence accuracy, in neural network models. Secondly, the consistency of the global model's performance across clients is assessed using violin plots; a flatter violin plot indicates a fairer global model. Next, baseline algorithms for personalized federated learning are compared to examine the improvements in global accuracy and individual fairness provided by FedMC and FedMC+. Ablation experiments on FedMC+ are then conducted to analyze the combined effects and contributions of the three mechanisms. Finally, the impact of fine-tuning the global gradient on individual performance is investigated.

5.1. Experimental setup

5.1.1. Data set

In the experimental design, all experimental studies are conducted on two image recognition datasets: (1) EMNIST (by letters) which covers 37 categories products with 60/10 thousand training test data divisions of 28 × 28 pixels grayscale images; (2) CIFAR10 containing 10 classes universal objects with 50/10 thousand training test data divisions of 32 × 32 pixels color images and (3) CIFAR100 containing 100 classes universal objects with 50/10 thousand training test data divisions of 32 × 32 pixels color images, each of which has two labels: fine-labels and coarse-labels.

5.1.2. Data distribution

In all experiments, the non-iid distribution is categorized into two types: (1) Pathological non-iid, wherein each client possesses a predetermined number of data categories employed for the partitioning of CIFAR10 and CIFAR100 datasets; and (2) Empirical non-iid, generated by sampling from the Dirichlet distribution [44], closely mirroring real-world non-iid scenarios which is applied to segment EMNIST dataset.

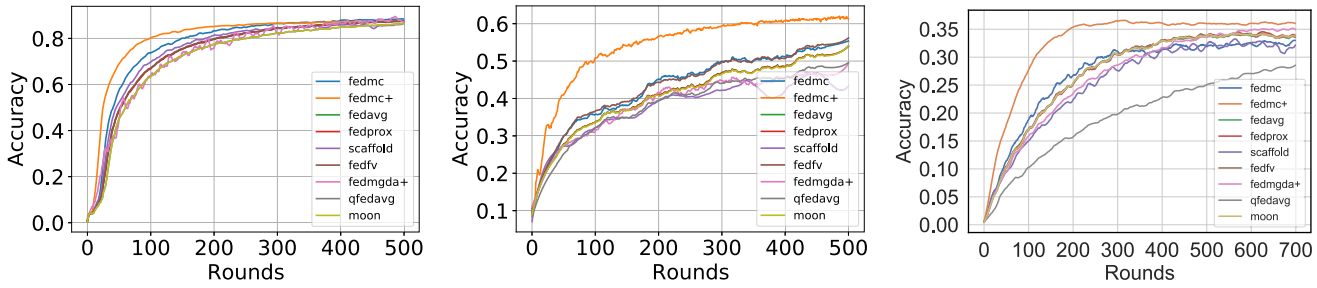


Fig. 3. Comparison of global model accuracy of each algorithm. (a) is trained on the EMNIST dataset; (b) is trained on the CIFAR10 dataset; (c) is trained on the CIFAR100 dataset.

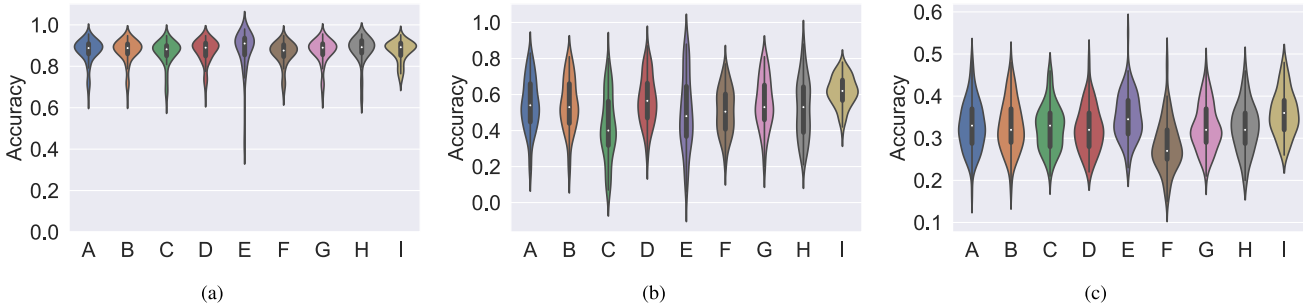


Fig. 4. Comparison of global model fairness of each algorithm. (a) is trained on the EMNIST dataset (A-fedavg, B-fedprox, C-scaffold, D-fedfv, E-fedmgda+, F-qfedavg, G-moon, H-fedmc, I-fedmc+); (b) is trained on the CIFAR10 dataset; (c) is trained on the CIFAR100 dataset.

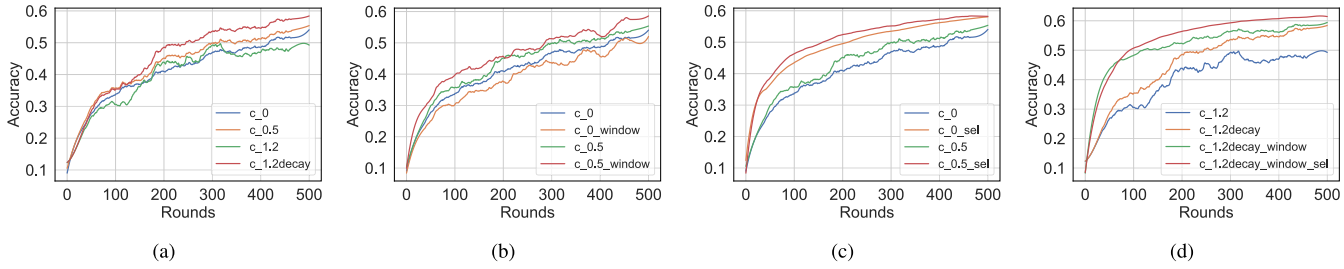


Fig. 5. The ablation experiment on the effect of three mechanisms on the algorithm. (a) is the decay of parameter c; (b) is the sliding window mechanism; (c) is the data-balanced client selection; (d) is the effect of the three mechanisms gradually superimposed.

5.1.3. Baseline algorithms

In our experiments for one-model FL framework, we compare several algorithms, including Fedavg [10], a classical federated learning algorithm, Fedprox [26], Scaffold [24], and Moon [32] which are designed to address the challenges posed by model performance degradation in non-iid scenarios and qfedavg [19], Fedmgda+ [20], Fedfv [22] to mitigate fairness issues arising from uneven performance of the global model in such cases.

In the framework of personalized federated learning, we select FedFomo [37], which also employs convex optimization problems to determine optimal weights for personalized model aggregation; Ditto [21], which is also a personalized algorithm to solve fairness problem; FedALA [42], a lightweight algorithm designed for adaptively updating the aggregate weights of personalized models.

5.1.4. Hyperparameter and model

In the experiment, the number of local rounds is fixed as $E = 1$, and the local learning rate is fixed as $\eta_l = 0.005$. In the local update, we use $B_{EMNSIT} = 50$, $B_{CIFAR10} = 10$ and $B_{CIFAR100} = 5$ as the batchsize for corresponding dataset of mini-batch gradient descent (MBGD). For the setting of the federated learning framework, we set the total number of clients $N = 100$, the proportion of selected clients in each round $C = 0.1$ and the global learning rate $\eta_g = 0.005$. For the neural network, CIFAR10 is trained on a seven-layer convolutional neural network, while EMNIST is trained on another seven-layer convolutional neural network [45].

5.2. Main results

In the experiment, we evaluate the performance of FedMC, FedMC+ and other baseline algorithms comprehensively. Firstly, the basic performance of the algorithm is evaluated by the accuracy of the global model on the test set. Secondly, the fairness of the global model is evaluated by the degree of accuracy dispersion on each client verification set. Thirdly, the effects of each mechanism on FedMC are evaluated by ablation experiments. Next, we classify and statistically analyze the performance of the global model on the individual client dataset by comparing and analyzing the average performance trend of different groups of clients before and after fine-tuning the global gradient. Finally, we compare the accuracy and accuracy variance of the global model in the personalized federated learning algorithm to illustrate the difference between the proposed algorithms and the personalized algorithms in performance and fairness improvement.

Table 1
Comparison of convergence rate and convergence loss of each algorithm. The R_a columns show the minimum number of rounds required to reach $a * 100\%$ of the accuracy in the experiment. The LOSS columns show the convergence loss value in the experiment.

Method	EMNIST			CIFAR10			CIFAR100		
	$R_{0.4}$	$R_{0.7}$	LOSS	$R_{0.3}$	$R_{0.4}$	LOSS	$R_{0.2}$	$R_{0.3}$	LOSS
Fedavg	41	111	0.362	45	98	1.202	119	251	1.073
Fedprox	41	111	0.363	45	103	1.203	122	261	1.099
Scaffold	37	99	0.397	41	191	1.515	150	318	2.246
Fedmgda+	38	115	0.314	41	110	2.165	140	316	1.692
qfedavg	46	127	0.416	61	143	1.337	272	/	2.740
Fedfv	37	111	0.361	42	90	1.145	118	262	1.064
Moon	41	111	0.362	52	103	1.189	122	262	1.045
FedMC	33	83	0.318	42	75	1.207	97	218	**0.724**
FedMC+	**22**	**53**	0.382	**21**	**44**	**0.963**	**59**	**134**	0.801

5.2.1. Comparison of model performance

Fig. 3 shows the trend of test accuracy of each algorithm on CIFAR10, CIFAR100, and EMNIST datasets. On the EMNIST dataset, each algorithm has a relatively high convergence accuracy compared to the extremely non-iid data distribution due to the fact that almost every client has more categories of data. As depicted in Fig. 3(a), the accuracy achieved by each algorithm on the test set upon final convergence is as follows: Fedavg: 0.878; Fedprox: 0.877; Scaffold: 0.871; Moon: 0.877; qfedavg: 0.865; Fedmgda+: 0.873; Fedfv: 0.876; FedMC: 0.885; FedMC+: 0.874. Furthermore, FedMC+ and FedMC also show faster convergence rates.

In CIFAR10, FedMC+ shows higher convergence accuracy and faster convergence rate. However, FedMC, which does not include the three mechanisms, has limited improvement in model performance in federated learning. It can also be observed from Fig. 3(b) that the accuracy of the test set achieved by each algorithm upon final convergence is as follows: Fedavg: 0.541; Fedprox: 0.539; Scaffold: 0.433; Moon: 0.541; qfedavg: 0.496; Fedmgda+: 0.493; Fedfv: 0.562; FedMC: 0.554; FedMC+: 0.614.

In CIFAR100, FedMC+ and FedMC also have similar performance as in CIFAR10. As evident from Fig. 3(c), each algorithm achieves the following test accuracy, respectively: Fedavg: 0.337; Fedprox: 0.338; Scaffold: 0.323; Moon: 0.337; qfedavg: 0.286; Fedmgda+: 0.348; Fedfv: 0.335; FedMC: 0.329; FedMC+: 0.360. The three added mechanisms designed to enhance FedMC have notably improved the performance of its global model. As presented in Table 1, we provide a quantitative indicator of convergence rate, specifically the minimum number of rounds required for each algorithm to achieve a specific test accuracy. Additionally, we display the convergence loss value of each algorithm, calculated as the average of the loss values from the last five rounds. The data shows the relatively faster convergence rate of FedMC+ and the relatively lower convergence loss value of FedMC and FedMC+.

5.2.2. Comparison of fairness

We characterize fairness by examining the accuracy distribution of the last round of the global model on each client's validation set. As shown in Fig. 4, we leverage the violin plot to characterize the dispersion of the accuracy of each client validation set. Among them, FedMC+ violin plot is more clustered, indicating that the global model is more uniformly distributed on each client verification set. The more consistent distribution shows that the global model has better adaptability on each client data set. In addition to the consistency of the global model, the higher the overall distribution of the violin plot, the better the performance of the global model.

As an indicator of the consistency in the performance distribution of the global model, we can utilize the variance of the accuracy of the global model on each client's validation set where a lower variance suggests a tighter distribution. In federated learning, the pursuit of global model is a non-single objective. We aim to obtain a global model with better average performance, as well as a global model with more consistent or equitable performance across local clients. In this context, our expectation is for the global model to exhibit higher average performance per local client and less variance in performance across local clients. Here, we employ the dispersion coefficient to quantify the degree of dispersion in each client's accuracy distribution. Its calculation is to divide the standard deviation by the average value. This metric is effective in eliminating the influence of measurement scale and dimension, providing a clearer reflection of the dispersion degree in a set of data. In the context of federated learning, smaller dispersion coefficients indicate that the global model exhibits better comprehensive performance. Table 2 presents the mean, standard deviation, and dispersion coefficient of the global model's validation set accuracy across clients. Notably, FedMC+ exhibits the smallest dispersion coefficient, suggesting superior overall performance for the algorithm.

On all datasets, FedMC+ also belongs to the position above the violin plot. FedMC+ can, therefore, improve the accuracy and consistency of the model at the same time. Due to the partial aggregation mechanism of federated learning, FedMC only solves the gradient conflict problem of partially selected clients, and this direction cannot be used as the direction to solve the global gradient conflict. FedMC+ can solve the global gradient conflict problem by using more local gradient information through a sliding window mechanism and a data-balanced client selection mechanism.

5.2.3. Ablation experiment

In the ablation experiment, we compare the individual enhancement provided by each mechanism in FedMC, as well as the combined effect achieved by incorporating all the mechanisms. As depicted in Fig. 5, we have plotted the individual effect of each mechanism on the algorithm as well as the combined effect achieved by their superposition. In the selection of c, we choose $c = 0.5$ and $c = 1.2$ respectively for comparison. At $c = 0.5$, there is a marginal increase relative to Fedavg ($c = 0$). However, when $c = 1.2$, there is a significant oscillation, and the model performance does not exhibit substantial improvement. When the decay mechanism is added to $c = 1.2$, the oscillation problem is effectively addressed, resulting in an improvement in model performance. From Fig. 5(a), it can be observed that when $c = 0$, the accuracy of convergence reaches 0.541. With $c = 0.5$, the accuracy of convergence improves to 0.554. However, when $c = 1.2$, the accuracy of convergence decreases to 0.492. Nevertheless, when $c = 1.2$ and the parameter decay mechanism is added, the accuracy of convergence improves to 0.584. The sliding window mechanism does not contribute to positive performance improvement for Fedavg. However, when applied to FedMC ($c = 0.5$), the sliding window mechanism expands the search space of FedMC, providing more information about the local gradient from a global perspective. From Fig. 5(b), it is evident that when $c = 0$, the accuracy of convergence reaches 0.541. Upon adding the sliding window mechanism to $c = 0$, the accuracy of convergence decreases to 0.521. When $c = 0.5$, the accuracy of convergence improves to 0.554. With the addition of the sliding window mechanism to $c = 0.5$, the accuracy of convergence further increases to 0.586.

Regarding the data-balanced client selection mechanism, it has a significant impact on enhancing the algorithm. From Fig. 5(c), it can be observed that when $c = 0$, the accuracy of convergence reaches 0.541. Upon adding data-balanced client selection to $c = 0$, the accuracy of convergence improves to 0.581. When $c = 0.5$, the accuracy of convergence reaches 0.554. With the addition of client selection to $c = 0.5$, the accuracy of convergence further increases to 0.582. It reduces convergence oscillation, improves convergence rate, and enhances accuracy. As the three mechanisms are gradually superimposed on FedMC, the improvement in model performance steadily increases. From Fig. 5(d), it is observed that when $c = 1.2$, the accuracy of convergence reaches 0.492. Upon adding the parameter decay mechanism at $c = 1.2$, the convergence accuracy improves to 0.584. Combining

Table 2
The table below presents the mean, standard deviation, and dispersion coefficient of the validation accuracy for each algorithm's global model across various clients within each dataset.

Method	EMNIST			CIFAR10			CIFAR100		
	MEAN	STD	STD/MEAN	MEAN	STD	STD/MEAN	MEAN	STD	STD/MEAN
Fedavg	0.873	0.060	0.069	0.543	0.146	0.268	0.329	0.059	0.178
Fedprox	0.872	0.060	0.068	0.544	0.145	0.267	0.330	0.060	0.183
Scaffold	0.863	0.064	0.075	0.421	0.180	0.429	0.326	0.054	0.165
Fedmgda+	0.877	0.105	0.120	0.499	0.207	0.414	0.351	0.055	0.158
qfedavg	0.860	0.059	0.069	0.507	0.128	0.253	0.283	0.060	0.212
Fedfv	0.872	0.060	0.069	0.567	0.137	0.241	0.322	0.054	0.169
Moon	0.872	0.062	0.071	0.549	0.144	0.263	0.326	0.054	0.167
FedMC	**0.880**	0.066	0.076	0.523	0.164	0.313	0.325	0.058	0.178
FedMC+	0.873	**0.051**	**0.059**	**0.617**	**0.085**	**0.137**	**0.360**	**0.053**	**0.148**

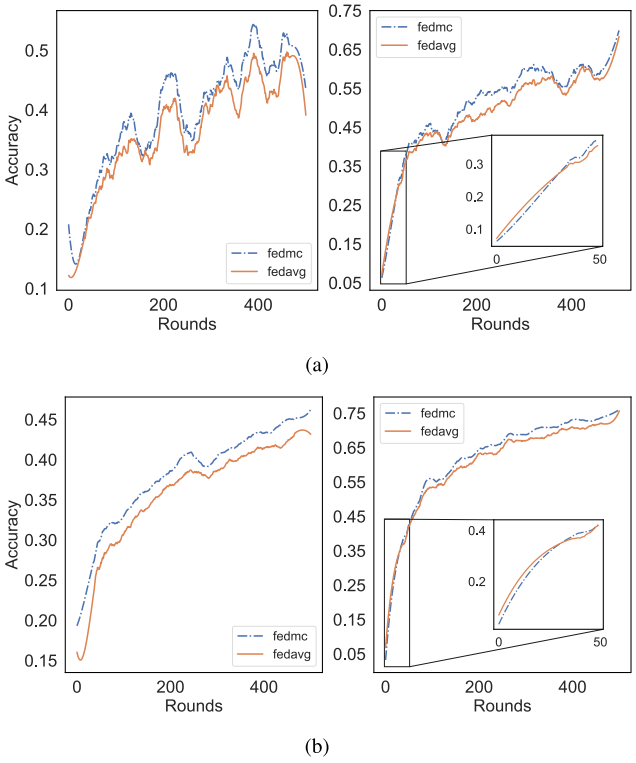


Fig. 6. The average performance trend of the global model on the bottom 10% of the worst-performing clients and the top 10% of the best-performing clients. (a) the client selection ratio is 10%; (b) the client selection ratio is 100%.

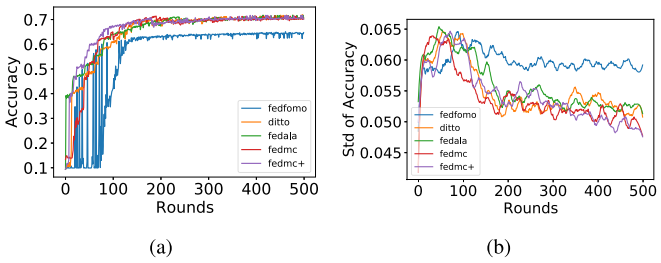


Fig. 7. Comparison with the global model of personalized federated learning algorithm. (a) Accuracy curve with the rounds; (b) Std of Accuracy curve with the rounds.

$c = 1.2$ with the parameter decay mechanism and the sliding window mechanism further increases the accuracy to 0.594. Finally, when $c = 1.2$ is combined with the parameter decay mechanism, sliding window mechanism, and data-balanced client selection, the accuracy reaches 0.614 during convergence.

5.2.4. Performance on individual clients

In this subsection, we illustrate performance trends for individual clients and elucidate the mechanism of fine-tuning global model in the FedMC algorithm. This is achieved by comparing clients in the top 10% of performance with those in the bottom 10%, as depicted in Fig. 6. We consider two scenarios for client selection ratios: one where client participation is $C = 0.1$ per round and another where client participation is $C = 1$ per round. We take the average of the top 10% performance of the global model on the local dataset to represent the high-performing clients, and likewise, we take the average of the bottom 10% performance to represent the low-performing clients.

As illustrated in Fig. 6, individual client performance trends exhibit more fluctuations when certain clients participate in each round of aggregation.

However, varying client participation rates exhibit similar performance trends across different individual clients. For clients with poor performance, the fine-tuning of the global model in the FedMC algorithm can be beneficial for their performance improvement, as shown in the two diagrams on the left of Fig. 6 where FedMC curves surpass those of Fedavg. While for clients with good performance, the FedMC algorithm may sacrifice its convergence rate in the early stages. However, in the later stages, the FedMC algorithm enables these high-performing clients to converge rapidly and achieve the desired accuracy level without fine-tuning the global model.

From the experiment, we can see that after fine-tuning the global gradient, the angle between the global gradient and the local gradients with good performance will be larger because the global gradient will be deflected to those local gradients with poor performance, which will lead to the performance of the global model on the data set of those local clients with good performance will be worse. However, the degree of reduction is relatively small, which indicates that the angle between the global gradient and those local gradients with good performance is still very likely to be acute. Therefore, as long as the training time is long enough, these clients with good performance will still achieve the convergence accuracy, but the convergence rate will be reduced in the earlier stage. At the end of the training, the convergence rate of the client with good performance will increase because in the process of fine tuning, the global gradient may be given the ability to jump out of the local optimal point.

5.2.5. Comparison with personalized FL

Since the framework of personalized federated learning is not proposed in this paper, in order to compare under the same standard, the personalized algorithms compared in this experiment all aggregate a global model by weighted average in the server each round. The evaluation of the experimental results is to measure the relevant indicators

of the global model. As shown in Fig. 7, in the comparison experiment of personalized federated learning, we adopt the setting of Dirichlet non-iid with $\alpha = 0.1$ for the CIFAR10 dataset. We all set the selected ratio for each round is $C = 0.1$.

In the comparison of accuracy, the proposed algorithm in the paper has a slight advantage in the convergence rate, but it achieves almost the same convergence accuracy as other personalized algorithms in the final convergence. However, in the comparison of standard deviation of accuracy, the proposed algorithm can have a lower standard deviation. This shows that our proposed algorithm has better performance in fairness and can also produce a global model with stronger consistency in model performance distribution.

6. Conclusion

In this paper, we formulate a multi-objective optimization approach within the context of federated learning to holistically address the challenges stemming from the degradation of global model performance and fairness issues arising from non-iid data distribution. We introduce FedMC as a solution targeting gradient conflicts, formulating a convex optimization problem to directly mitigate disparities between the global gradient and local gradients, thereby obtaining a fine-tuned global gradient. Additionally, we present FedMC+, an enhanced version of FedMC, incorporating three optimization mechanisms: hyperparameter decay, sliding window mechanism, and data-balanced client selection. Theoretical analysis is conducted to examine the convergence rate of FedMC and demonstrate its convergence to a Pareto stationary point. Extensive experiments have demonstrated the superior efficiency, accuracy, and fairness of FedMC+ compared to other state-of-the-art federated learning algorithms. This work places a particular emphasis on the multi-objective demands of the global model in federated learning under the influence of non-iid data. It seeks to ensure that the performance distribution of the global model on each local dataset is as even as possible without compromising overall global performance.

CRediT authorship contribution statement

Yuhao Shen: Writing – original draft, Software, Methodology, Conceptualization. **Wei Xi:** Writing – original draft, Validation, Supervision. **Yunyun Cai:** Validation, Data curation. **Yuwei Fan:** Validation, Data curation. **He Yang:** Validation, Data curation. **Jizhong Zhao:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declared that they have no conflicts of interest to work.

Data availability

The data used in this article are all public datasets.

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Appendix A. Proof of theorems

A.1. Proof of Theorem 1

Proof. Consider the tth optimization step and denote g_t^* the update direction obtained by solving (2) at the tth iteration. Then

$$
\begin{aligned}
&\mathbb{E}[F(w_{t+1}) - F(w_t)] \\
&\le -\alpha \nabla^T F(w_t) \cdot \mathbb{E} g_t^* + \frac{L\alpha^2}{2} \mathbb{E}\|g_t^*\|^2 \\
&\le -\alpha \nabla^T F(w_t) \cdot \mathbb{E} g_t^* + \frac{\alpha}{2} \mathbb{E}\|g_t^*\|^2 \\
&\le -\frac{\alpha}{2}(\|\nabla F(w_t)\|^2 - \mathbb{E}\|\nabla F(w_t) - g_t^*\|^2 + \mathbb{E}\|g_t^*\|^2) + \frac{\alpha}{2}\mathbb{E}\|g_t^*\|^2 \\
&= -\frac{\alpha}{2}(\|\nabla F(w_t)\|^2 - \mathbb{E}\|\nabla F(w_t) - g_t^*\|^2) \\
&\le -\frac{\alpha}{2}[\|\nabla F(w_t)\|^2 - \mathbb{E}(\|\nabla F(w_t) - \bar{g}_t\| + \|\bar{g}_t - g_t^*\|)^2] \\
&\le -\frac{\alpha}{2}[\|\nabla F(w_t)\|^2 - (\mathbb{E}\|\nabla F(w_t) - \bar{g}_t\|^2 + \mathbb{E}\|\bar{g}_t - g_t^*\|^2 \\
&\qquad + 2\mathbb{E}\|\nabla F(w_t) - \bar{g}_t\| \cdot \|\bar{g}_t - g_t^*\|)] \\
&\overset{(A)}{\le} -\frac{\alpha}{2}[\|\nabla F(w_t)\|^2 - (\sigma^2 + c^2\mathbb{E}\|\bar{g}_t\|^2 \\
&\qquad + 2\sqrt{\mathbb{E}\|\nabla F(w_t) - \bar{g}_t\|^2 \cdot \mathbb{E}\|\bar{g}_t - g_t^*\|^2})] \\
&\le -\frac{\alpha}{2}[\|\nabla F(w_t)\|^2 - (\sigma^2 + 2c\sigma\sqrt{\mathbb{E}\|\bar{g}_t\|^2} + c^2\mathbb{E}\|\bar{g}_t\|^2)] \\
&\le -\frac{\alpha}{2}[\|\nabla F(w_t)\|^2 - (\sigma^2 + 2c\sigma G + c^2\sigma^2 + c^2\|\nabla F(w)\|^2)] \\
&\le -\frac{\alpha}{2}(1 - c^2)\|\nabla F(w_t)\|^2 + \frac{\alpha(\sigma^2 + 2c\sigma G + c^2\sigma^2)}{2},
\end{aligned}
$$

where (A) applies Cauchy-Schwartz equality. Using telescoping sums, we have

$$
\begin{aligned}
F(w_{T+1}) - F(w_0) = &-\frac{\alpha(1 - c^2)}{2} \sum_{t=0}^{T} \|\nabla F(w_t)\|^2 \\
&+ \frac{\alpha(T+1)(\sigma^2 + 2c\sigma G + c^2\sigma^2)}{2}.
\end{aligned}
$$

Therefore,

$$
\frac{1}{T+1}\sum_{t=0}^{T}\|\nabla F(w_t)\|^2 \le \frac{2(F(w_0) - F(w_{T+1}))}{\alpha(1-c^2)(T+1)} + \frac{\sigma^2 + 2c\sigma G + c^2\sigma^2}{1 - c^2}. \quad \square
$$

A.2. Proof of Theorem 2

Proof. For each task, we have $i \in [N]$:

$$
\begin{aligned}
&f_i(w_{t+1}, \xi) - f_i(w_t, \xi) \\
&\le -\alpha_t \nabla^T f_i(w_t, \xi) \cdot g_t^* + \frac{L_1\alpha_t^2}{2}\|g_t^*\|^2 \\
&\le -\alpha_t \min_i \nabla^T f_i(w_t, \xi) \cdot g_t^* + \frac{L_1\alpha_t^2}{2}\|g_t^*\|^2 \\
&\le -\alpha_t ({g_{p^*}^t}^T \cdot \bar{g}_t + c\|\bar{g}_t\| \cdot \|g_{p^*}^t\|) + \frac{L_1\alpha_t^2}{2}\|g_t^*\|^2,
\end{aligned}
$$

where the last step of the inequality applies (7). Meanwhile, according to the expression of g_t^*, we have

$$
\begin{aligned}
\|g_t^*\|^2 &= \|\bar{g}_t + c\frac{\|\bar{g}_t\|}{\|g_{p_t^*}\|} g_{p_t^*}\|^2 \\
&= (1 + c^2)\|\bar{g}_t\|^2 + 2c\frac{\|\bar{g}_t\|}{\|g_{p_t^*}\|} {g_{p_t^*}}^T \cdot \bar{g}_t \\
&= 2c\frac{\|\bar{g}_t\|}{\|g_{p_t^*}\|}({g_{p_t^*}}^T \cdot \bar{g}_t + c\|\bar{g}_t\| \cdot \|g_{p_t^*}\|) + (1 - c^2)\|\bar{g}_t\|^2.
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
&f_i(w_{t+1},\xi) - f_i(w_t,\xi) \\
&\le -\alpha_t(1 - L_1\alpha_t c\frac{\|\bar{g}_t\|}{\|g_{p_t^*}\|}) \cdot ({g_{p_t^*}}^T \cdot \bar{g}_t \\
&\quad + c\|\bar{g}_t\| \cdot \|g_{p_t^*}\|) - \frac{L_1\alpha_t^2}{2}(c^2-1)\|\bar{g}_t\|^2 \\
&\overset{(A)}{\le} -\alpha_t(1 - L_1\alpha_t c\frac{\|\bar{g}_t\|}{\|g_{p_t^*}\|}) \cdot (c-1)\|\bar{g}_t\| \cdot \|g_{p_t^*}\| \\
&\quad - \frac{L_1\alpha_t^2}{2}(c^2-1)\|\bar{g}_t\|^2 \\
&= -\alpha_t(c-1)\|\bar{g}_t\| \cdot \|g_{p_t^*}\| + L_1\alpha_t^2 c(c-1)\|\bar{g}_t\|^2 \\
&\quad - \frac{L_1\alpha_t^2}{2}(c^2-1)\|\bar{g}_t\|^2 \\
&\le -\alpha_t(c-1)\|\bar{g}_t\| \cdot \|g_{p_t^*}\| + \frac{L_1\alpha_t^2}{2}(c-1)^2\|\bar{g}_t\|^2 \\
&\overset{(B)}{\le} -\frac{1}{2}\alpha_t(c-1)\|\bar{g}_t\| \cdot \|g_{p_t^*}\|,
\end{aligned}
$$

where (A) applies Cauchy–Schwarz inequality and (B) uses $\alpha_t \le \frac{\|g_{p_t^*}\|}{L_1(c-1)\|\bar{g}_t\|}$. Take the telescoping sum on both sides of this inequality, we have:

$$
\sum_{t=0}^{T} \alpha_t\|\bar{g}_t\| \cdot \|g_{p^*}^t\| \le 2\frac{min_i(f_i(w_0) - f_i(w_{T+1}))}{c-1},
$$

where $c \ge 1$. Let $\alpha_t = \frac{\|\bar{g}_t\|}{L_1(c-1)\|g_{p_t^*}\|}$, we have

$$
\sum_{t=0}^{T} \|\bar{g}_t\|^2 \le 2L_1 min_i(f_i(w_0) - f_i(w_{T+1})).
$$

Take the expectation on both sides,

$$
\begin{aligned}
\sum_{t=0}^{T} \|\nabla F(w)\|^2 &= \sum_{t=0}^{T} \|\mathbb{E}\bar{g}_t\|^2 \\
&\le \sum_{t=0}^{T} \mathbb{E}\|\bar{g}_t\|^2 \le 2L_1 min_i(f_i(w_0) - f_i(w_{T+1})),
\end{aligned}
$$

which implies $\|\nabla F(w)\|^2 \to 0$. □

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